

Homework 01

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Question 1: You are going to learn how to store structured data on a computer system by using database approach. For example, you will record employee information and their working process at a company by using data models implemented in DBMS.

Suppose that companies want to store employee's photo, such as the whole face or fingerprint or iris, for identification purposes. Let's discuss this data type. How to store them in a database in order to search/analyze effectively? Do we need another kind of data systems to manipulate with them?

Companies also have to store employment contracts. Is the database approach for photos still suitable for document data or not? Support your answer and suggest an appropriate DBMS for this type of data.

Answer:

Discuss the data type:

- Employee photos: This is a standard image file format (JPEG, PNG) containing the employee's facial data.
- Fingerprint: Fingerprint data can be stored in different formats, which are often unique to the device used to capture them [1]. It usually includes the details of the fingerprint, such as where the ridges end and split [2].
- Iris scan: This device takes a picture of the unique pattern in the eye's iris and store them in digital format [3].

These data involve with the biometric data type which refer to the authentication and identification [4]. Since these digital images is represented as binary data, biometric data are typically stored as BLOBs (Binary Large Object) in database [5]. BLOBs are used to stored large binary data which include images, audio, and video files. These data types are suitable for storing raw image data and ensure that the integrity and fidelity of the image are maintained.

Concerns:

- Privacy Issues: Storing biometric data raises privacy concerns [6]. Employees may be uncomfortable with their unique identifiers being collected and stored.
- Data Security Risks: If biometric data is stolen, it can't be changed like a password [7]. To reduce these risks, strong security measures are needed.

How to store them in a database in order to search/analyze effectively?

There are two main ways to store images in database using traditional DBMS: using file systems or object storage services [5]. But I recommend using the object storage services to gain high speed and reduces the size of the photo it will load in the database.

Here's the short demo structure of our table:

```
CREATE TABLE image_path_table (
```

```
id SERIAL PRIMARY KEY,  
name VARCHAR(255),  
image_path TEXT  
); [5]
```

This way is straight forward to reference the location of the image file within your database, but you can let your web server or a third-party service (such as Amazon S3, Google Cloud Storage) serve the image directly [5]. Once you upload your image on this server, you can get the URLs of the picture.

Benefits by using this method:

- Reduced database size: Images can take up significant storage space. Storing only the path keeps the database smaller and potentially reduces storage costs.
- Simpler database management: Database backups and maintenance become faster and easier when dealing with smaller data sizes.

Do we need another kind of data systems to manipulate with them?

I have done a research on the Internet and I find a system called Haystack in the article "Needle in a Haystack: Efficient Storage of Billions of Photos" [8]. In this blog, the writers discusses Facebook's infrastructure challenges in storing billions of photos and their solution called Haystack.

- Minimized Metadata Overhead: Traditional file systems store a lot of metadata (such as file attributes and directory structures) which can slow down access times. Haystack reduces this overhead by combining photo data and its associated metadata into a single object, make it more easier for retrieval and storage.
- Efficient Indexing: Haystack uses an efficient indexing system that maps photo IDs directly to their locations on disk. This direct mapping allows for faster lookup times compared to traditional hierarchical file systems.
- Optimized Storage: By reducing the number of disk operations required to store and retrieve photos, Haystack improves the overall performance and efficiency of the storage system.

Is the database approach for photos still suitable for document data or not?

I think in this case, the method that I use for storing images in database cannot apply for document data for this reason:

- Document data such as employment contracts usually contains structure of text, which refers to data type like text, varchar. Storing the file path of the contract is not a suitable approaches since we need to handle lots of modification along the contract life time. For biometric data type, it is rarely to overwrite and modify the information of that data since this data used only for authentication and identification. A contract is more than that, it involves transaction, sign contract between industries and therefore we need to store the actual data in database, instead of reference to the original file the storing images.

For handling documents such as employment contracts, databases designed for document storage, retrieval, and analysis might be more appropriate.

I would go for PostgreSQL:

- PostgreSQL is known for its strong data integrity and reliability. This ensures that your employment contracts are stored safely and can be accessed without errors or data corruption.[12]
- PostgreSQL offers advanced features like full-text search, JSON support, and complex queries. These features make it easier to search, manage, and analyze contract data efficiently.[12][13]
- PostgreSQL can handle a large volume of data and users. As your business grows and the number of employment contracts increases, PostgreSQL can scale to meet your needs without compromising performance.[12]

Question 2: From Facebook's applications, such as website and mobile app, let's take a guess: What types of data does Facebook manage? For each type, which DBMS does Facebook use? and why does Facebook use that DBMS?

Facebook manages a vast array of data types from its various applications, including both explicit user-provided data and implicit data inferred from user behavior. Here are some key types of data that Facebook manages:

Type of data that Facebook would manage:

1. Personal information:
 - Name, Age, Gender: Basic user profile information.
 - Contact Information: Email addresses, phone numbers.
 - Birthday, Location: Date of birth, current city, hometown.
2. Social data:
 - Friends and Connections: Lists of friends, groups, pages followed.
 - Family Relationships: Information on family members.
 - Events: Invitations to and participation in events.
3. Content data:
 - Posts and Status Updates: Text, images, videos, and links shared by users.
 - Comments and Likes: User interactions with posts, including reactions and comments.
 - Shared Media: Photos, videos, live streams.
4. Interaction data:
 - Messages: Private messages and chat history through Messenger.

- Interactions: Likes, reactions, comments, shares, and other engagement metrics.
 - Page Interactions: Interactions with public pages and businesses.
5. Behavioral data:
 - Browsing Activity: Pages visited, content viewed, time spent on different parts of the site.
 - App Usage: Frequency and duration of app usage.
 - Clicks and Navigation: Patterns of how users navigate through the app and website.
 6. Location data:
 - Check-ins: Locations where users check in.
 - GPS Data: Location data from mobile devices.
 7. Device information:
 - Device Types: Information about the devices used to access Facebook.
 - IP Addresses: Internet Protocol addresses logged when users access Facebook.
 - Cookies: Data collected through cookies and similar technologies.
 8. Advertising and Purchase data:
 - Ad Data on which ads users see, click, and interact with.
 - Purchase History: Data on transactions and purchases made through Facebook.
 9. Security information:
 - Login History: Logs of login times, locations, and devices.
 - Security Settings: Two-factor authentication, security questions, and responses.

Suggested DBMS for each type:

1. Personal information
 - DBMS: MySQL
 - Why? MySQL is a popular open-source relational DBMS known for its scalability, reliability, and ease of use [14]. It's efficient for storing structured data like names, birthdays, and locations.
2. Social data
 - DBMS: MySQL or Custom Sharded Solution
 - Why? Social data can involve complex relationships and connections [15]. While MySQL can handle basic friend lists, Facebook might use a custom sharding solution for scalability. Sharding involves dividing data across multiple servers based on specific criteria (e.g., user ID) for faster retrieval and updates [16].
3. Content data
 - DBMS:
 - o MySQL (text and metadata)
 - o NoSQL (like Cassandra) for large content like videos and photos

- Why? Content data has different needs. Textual content and metadata might store in MySQL, while large files like videos and photos might be stored in a NoSQL solution like Cassandra. NoSQL databases perform well at handling unstructured or large data volumes.
- 4. Interaction data
 - DBMS: HBase
 - Why? HBase prioritizes real-time access for both reading and writing messages[17]. Additionally, HBase's ability to handle large datasets efficiently ensures it can scale to accommodate Facebook's massive user base and message volume [18].
- 5. Behavioral data
 - Potential DBMS:
 - o Apache Hadoop
 - o Clickstream analytics databases
 - Why? Behavioral data involves complex analysis and requires different tools. Apache Hadoop, a distributed file system, can handle massive amounts of behavioral data for batch processing [19]. Specialized clickstream analytics databases can analyze user actions in real-time.
- 6. Location data
 - Potential DBMS: Geospatial databases (PostGIS)
 - Why? Location data requires spatial queries. Facebook might utilize PostGIS, a spatial extension for PostgreSQL, or similar solutions within MySQL to efficiently manage location check-ins and geospatial data. [20]
- 7. Device information
 - DBMS: MySQL
 - Why? Device information is structured data and could be stored in MySQL.[14]
- 8. Advertising and Purchase data
 - DBMS: MySQL
 - Why? This data involves transactions and historical analysis which can be analyzed and handled easily by MySQL [14]
- 9. Security information
 - DBMS: MySQL
 - Why? Security information requires high levels of protection. Facebook likely uses MySQL for login data and potentially specialized the security event [21] [22].

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